

# Archive-aware forecasting and category calibration for delayed conjunction reports

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## Abstract

Censored regression, two-part models, and decision-aware evaluation are established statistical tools. Their application to ESA’s Collision Avoidance Challenge requires an unresolved distinction: the archive’s dominant recorded value of  $-30$  has no documented generation rule in the public release. We forecast later reported log collision risk from messages available at least 48 hours before closest approach, treating  $-30$  as an observed atom rather than proven latent censoring. We compare direct and residual regression, a refitted existing hurdle, a calibrated mixture mean, and a guarded hurdle selected on validation challenge loss. On the original 1,659-event local split, the selected remedy reduces non-floor MAE from 4.073 to 3.309, while tying persistence at ESA loss 0.167. On 2,167 released official events, non-floor MAE falls from 4.287 to 3.192, with the same ESA loss 0.694. Outcome-category conformal inversion increases the direct regressor’s official high-risk coverage from 24.0% to 93.3%. This calibration remedy does not eliminate false reassurance. Controlled simulations test recording mechanisms and temporal drift; they are mechanism evidence, not cross-domain empirical validation. The contribution is a reproducible conjunction benchmark extension using known methods.

## 1 Problem and contribution boundary

ESA’s task is to predict the last reported base-10 log collision probability from earlier Conjunction Data Messages. The target is an updated estimate, not a collision occurrence. The published challenge already emphasizes high-risk loss and discusses the dominant  $-30$  mass and residual prediction [1]. Aggregate accuracy does not establish manoeuvre utility.

The revision adds a target-semantics audit, operationally selected remedy comparisons, and candidate-category prediction sets. It applies established methods instead of claiming that floors or tail-sensitive evaluation are new. The intended destination is aerospace. No pretrained temporal foundation model is tested, and there is no conclusion about scaling. One empirical archive plus simulation does not establish a cross-domain machine-learning method.

## 2 Known background and archive semantics

Detection and quantitation limits describe distinct measurement properties [3]. Tobit accounts for a known censoring boundary under a latent-outcome likelihood [4]. Two-part models separate a boundary mass from a continuous response [5]; zero-inflated count models use a different likelihood [6]. Utility-aware regression and prediction for downstream optimization precede this study [7, 8]. Split conformal provides marginal coverage under exchangeability [9]; Mondrian categories adapt calibration to specified groups [10]. These guarantees do not supply unrestricted conditional coverage or validity under shift [11].

The official Zenodo release is CC BY 4.0 [2]. Its raw anonymized table has 82,484 of 199,082 message rows at  $-30$ , no lower risks, and no missing risk cells. Its documentation does not describe clipping, a censoring limit, or a missing-update sentinel; no executable preprocessing code is released. A partial unique-feature match finds 57,439 training messages at  $-30$  unchanged in the released raw table. PRISM did not create the atom, but its origin could be numerical saturation, a negligible-risk convention, a reporting bound, or another computation/export rule. We model the recorded value without inferring a latent probability of exactly  $10^{-30}$  or below it.

### 3 Protocol and remedies

The submission is preserved at commit 091d7c49fa74, tag fmts-2026-submitted. New settings are in the revision configuration. This is a retrospective revision on previously inspected outcomes, not a preregistered fresh test.

The cohort retains events with a final message less than one day before closest approach, an available message at least two days before it, and at least two messages. Rebuilding features gives the same 8,293 eligible events and event-disjoint split: 3,731 train, 1,659 validation, 1,244 calibration, and 1,659 test. All messages of an event share one split. Features use only messages at least two days before closest approach. The target is the later final report. Calendar times and object-pair IDs are absent even in the released raw table. Random event IDs are not time. Official labels supply an enriched 2,167-event transfer population; fitting uses only local training, validation, and calibration.

Let  $r$  be the latest available report,  $y$  the final report, and  $a = -30$ . Persistence predicts  $r$ . Direct XGBoost predicts  $y$ ; residual XGBoost predicts  $y - r$ . The refitted prior hurdle predicts  $a$  above a validation-MAE-selected atom threshold, otherwise using a residual regressor trained on non-atom targets. Its classifier retains the prior class weighting. Sigmoid calibration on validation corrects the probability scale, producing  $p(x)$  and mixture mean

$$\hat{y}_{\text{mix}} = p(x)a + (1 - p(x))\hat{y}_{\text{nonatom}}.$$

The revised grid combines calibrated hard hurdles at seven thresholds with five blends toward persistence, with or without a guard copying  $r$  when  $r \geq -6$ . Selection minimizes validation ESA loss, then decision cost, then non-floor MAE. This is a finite policy search, not a new decision-focused training algorithm. Tests never choose parameters.

Regressors use 250 trees, depth 4, learning rate 0.05, child weight 6, subsample 0.85, feature subsample 0.8, and regularization 0.1/2. The classifier uses 180 depth-3 trees, learning rate 0.05, child weight 5, and the same subsampling. Seeds 42–46 define event redraws. Runtime versions, hashes, feature names, and all candidate validation metrics are released. These models are refitted with the recorded runtime; their scores do not replace the submission’s frozen predictions.

#### 3.1 Metrics and uncertainty

The primary point metric is official loss  $L = \text{MSE}_{y \geq -6} / F_2$ . Forecasts below  $-6$  are clipped to  $-6.001$  for challenge scoring. If  $F_2 = 0$ , loss is infinite; if no positive target is present, it is undefined. JSON stores null and the reason instead of an artificial epsilon. Precision, recall, high-risk MSE, and floor/non-floor errors remain visible. For errors  $e_i$ , the additive population-weighted decomposition is

$$\text{MAE} = \frac{1}{n} \sum_{y_i=a} |e_i| + \frac{1}{n} \sum_{y_i \neq a} |e_i|.$$

These contributions are not a sum of stratum means.

Paired uncertainty uses 1,000 shared event bootstrap draws. Undefined ESA draws are counted and omitted from that loss interval. Stratum intervals resample within each stratum. Intervals are percentile intervals; redraw standard deviations are sensitivity summaries. Atom probabilities are scored by Brier loss. A decision proxy assigns false negatives 20, false positives 1, and review 0.25. These are analyst assumptions, not manoeuvre costs; review is optimistically assumed to resolve classification correctly.

#### 3.2 Outcome-category conformal inversion

Marginal split conformal uses one absolute-error radius. The category remedy computes separate corrected order-statistic radii on atom, non-floor low-risk, and high-risk calibration observations. For candidate  $z$ ,

$$z \in C(x) \iff |z - \hat{y}(x)| \leq q_{g(z)}.$$

The domains are  $\{a\}$ ,  $(a, -6)$ , and  $[-6, 0]$ . The result is a union of an optional atom and up to two intervals. The unseen actual category never chooses a radius. Absent or undersized categories receive infinite radii. Exchangeability within categories gives the usual category-conditional split-conformal rank guarantee; no new theorem is claimed.

Empty sets or sets containing both low- and high-risk possibilities require review. Width is the union’s length, not its convex hull. Marginal bands retain symmetric, unbounded-support width, so the two width summaries

are not equivalent; category sets exploit the specified recorded support. Accepted-set coverage has no conformal guarantee. Exact binomial intervals accompany coverage. Only nine local test events and eleven calibration events are high risk.

## 4 Measured remedy behavior

Seed 42 selects calibrated atom threshold 0.95, full hurdle weight, and the persistence guard. Table 1 leads with challenge loss. Mixture and prior hurdle have no local high-risk recall. The revised policy preserves persistence’s high-risk calls and clipped challenge MSE while improving non-floor error. Equal clipped score does not imply equal raw high-risk forecasts.

Table 1: Refitted local test: 1,659 events, 1,352 at  $-30$ , nine high risk. Error columns are stratum MAEs. Precision without positive calls is undefined.

| Model           | $L$      | Recall | Precision | Floor | Non-floor | MAE   |
|-----------------|----------|--------|-----------|-------|-----------|-------|
| persistence     | 0.167    | 0.778  | 0.115     | 5.308 | 4.073     | 5.080 |
| direct          | $\infty$ | 0.000  | –         | 1.724 | 7.682     | 2.826 |
| residual        | $\infty$ | 0.000  | –         | 1.717 | 7.454     | 2.778 |
| mixture mean    | $\infty$ | 0.000  | –         | 1.468 | 8.315     | 2.735 |
| old hurdle      | $\infty$ | 0.000  | –         | 0.461 | 9.793     | 2.188 |
| decision hurdle | 0.167    | 0.778  | 0.115     | 4.798 | 3.309     | 4.523 |

The selected remedy lowers overall MAE from 5.080 to 4.523, paired model-minus-persistence difference  $-0.557$   $[-0.771, -0.342]$ . Non-floor error falls from 4.073 to 3.309, difference  $-0.764$   $[-1.397, -0.193]$ . ESA loss stays 0.167, paired difference zero on all 998 valid draws; two draws have undefined loss. All five redraws select a guard, but they overlap and are not independent replications.

Official-transfer overall MAE is 4.352 versus 5.209, and non-floor MAE is 3.192 versus 4.287. Paired differences are  $-0.857$   $[-1.017, -0.704]$  and  $-1.095$   $[-1.550, -0.640]$ . ESA loss remains 0.694, recall 0.767, precision 0.646. Raw high-risk MAE difference is  $+0.267$   $[-0.295, 0.781]$ , so high-risk forecasting improvement is not established. Sigmoid calibration lowers local atom Brier loss from 0.1154 to 0.0876, paired difference  $-0.0279$   $[-0.0361, -0.0201]$ , but the mixture still misses every local high-risk event.

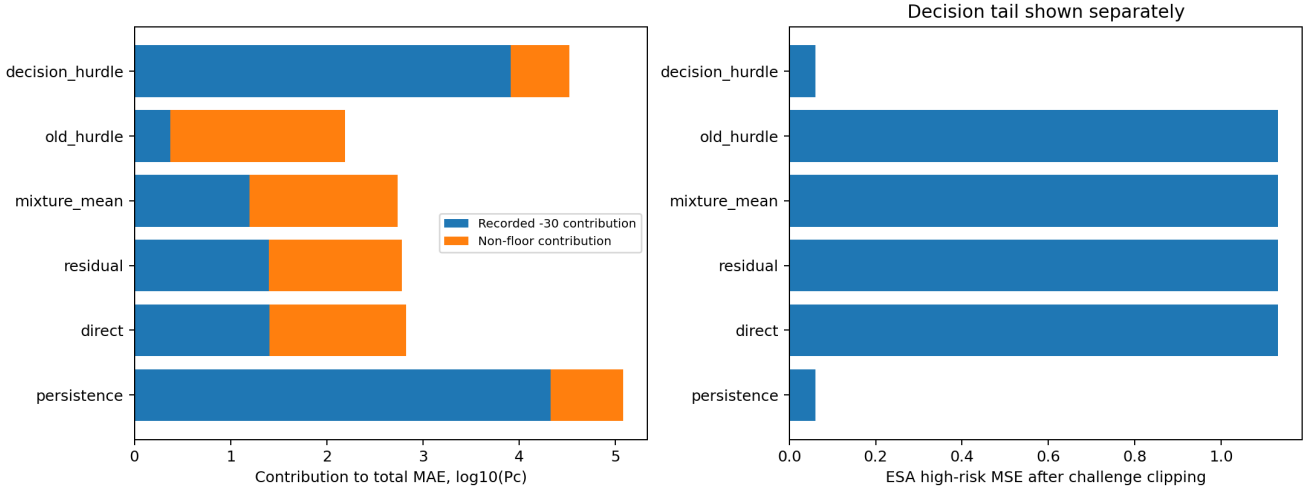


Figure 1: Floor and non-floor contributions sum to total MAE. A separate panel displays high-risk MSE after challenge clipping, keeping the decision tail visible.

Direct regression’s local high-risk coverage rises from 11.1% marginally to 88.9% with categories, based on nine events. Official coverage rises from 24.0% to 93.3%, the latter with exact interval  $[88.1, 96.8]\%$ . These are observed transfer outcomes, not shift guarantees.

The selected policy’s category sets cover 90.0% of official floor cases, 91.0% of non-floor low-risk cases, and 92.0% of high-risk cases. They review 30.0% and still accept 9 falsely reassuring high-risk forecasts. Mean

proxy cost is 0.158 versus 0.352 for persistence without review, but that comparison credits review with perfect resolution. Category sets around the refitted prior hurdle require review on every local and official event. Coverage alone does not establish useful abstention.

## 5 Controlled mechanism experiments

Sixty runs cross three recording mechanisms, mass rates 0.2/0.8, stationary/drifted variants, and five seeds. Each has 6,000 events, four preceding observations, known time, and ordered 45/20/15/20% event blocks. Predictor inputs precede targets; all observations of an event stay together. Gaussian Tobit is tested only under known clipping, with its latent prediction transformed to the expected recorded clipped response.

At stationary mass 0.8, Tobit has mean MAE 0.338 versus 0.374 for direct regression and 0.482 for the decision hurdle. This clipping regime has no high-risk test cases, so it offers measurement-mechanism evidence, not a decision comparison. With a structural atom at the same mass, mean MAE is 5.207 for direct regression and 10.177 for the decision hurdle, while their mean decision costs are 1.401 and 0.385. Error and cost prefer different policies under explicit simulated assumptions. Structural-atom and missing-update variants deliberately have identical observed data, illustrating semantic non-identifiability rather than independent replications. Drift changes target and mass propensity; observed coverage under drift is descriptive. None of this is a second empirical domain.

## 6 Limitations and declarations

The generation rule for  $-30$  requires producer documentation not found in the release. Calendar and object-pair validation are impossible with its identifiers. Official data are selected and enriched, not a chronological deployment sample. This retrospective study has already inspected historical test outcomes. Tail samples are scarce. A guarded model ties challenge loss through retained threshold calls and scoring clipping; it does not establish better raw tail forecasts. Conformal sets can review every case and have no shift guarantee. Costs omit manoeuvre constraints and reviewer errors. The deployed exhibit retains its historical model; these remedies are research outputs.

**Data and code.** ESA data and labels are available at DOI 10.5281/zenodo.4463683 under CC BY 4.0. PRISM code is MIT licensed at <https://github.com/ArjunCodess/PRISM>. This revision is local on fmts-review-improvements; it has not been pushed. Configurations, splits, hashes, predictions, intervals, failed remedies, tests, and generated tables accompany it. Simulation uses MIT-licensed code. The submitted manuscript remains in submissions/fmts.2026 and at the submission tag.

**Ethics.** Anonymized conjunction records involve no human-participant data. Forecasting later estimates does not validate collision outcomes or manoeuvre decisions. Remaining false reassurance precludes a safety claim. No agency endorsement is implied.

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**AI use.** OpenAI Codex assisted with source research, experiment and test code, execution, figures, tables, and manuscript revision at the author’s request. Source and numerical checks were performed against primary metadata and generated outputs during this session. This is not independent author verification. Author approval, verification, and any earlier AI-use history remain pending.

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